

AthenaK output for post-processing analysis

This document collects which AthenaK output variables are needed to perform a certain analysis in post-processing, e.g. the visualization of the evolution of kinetic and magnetic energy in the case of GRMHD simulations.

Ejecta and Poynting flux

VARIABLES

Metric: `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, `adm_gzz`

Primitives: `dens`, `s00`, `temperature`, `velx`, `vely`, `velz`, `bcc1`, `bcc2`, `bcc3`

Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`

Total: $N_{\text{vars}} = 23$

OUTPUT TYPE

```
<output1>
file_type = sph
variable = ... # see variable list
dt = 20.0
radius = 400.0 # radii = 300.0 400.0 ...
```

SETUP

Output time: 10M or 20M (the latter might already be too large)

Resolution: $(\theta, \phi)_{\text{low}} = (128, 256)$ or $(\theta, \phi)_{\text{high}} = (256, 512)$

Radii: Between 300M and 500M

DISK USAGE

- Low resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 23 \times 128 \times 256 \times 8 \approx 6.03\text{MB}(r\Delta t)^{-1}$
- High resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 23 \times 256 \times 512 \times 8 \approx 24.12\text{MB}(r\Delta t)^{-1}$

	10M	20M
• Total (100ms or 20000M):		
	Low resolution	High resolution
	$\approx 12.06 \text{ GBr}^{-1}$	$\approx 48.24 \text{ GBr}^{-1}$
	$\approx 6.03 \text{ GBr}^{-1}$	$\approx 24.12 \text{ GBr}^{-1}$

NOTE To save disk-usage, this can be paired with the output from **Neutrino luminosity and average energy density** and also **GRMHD butterfly diagrams**.

Neutrino luminosity and average energy density

VARIABLES

M1 variables: `rad_m1_F`, `rad_m1_E`, `rad_m1_N`

Metric: `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, `adm_gzz`

Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`

Total: $N_{\text{vars}} = N_{\text{vars,E/N}} \times N_{\text{species}} + \underbrace{N_{\text{vars,F}}}_{3 \text{ vector components}} \times N_{\text{species}} + N_{\text{vars,metric/gauge}} = 2 \times 4 + 3 \times 4 + 10 = 30$

OUTPUT TYPE

```
<output2>
file_type = sph
variable = ... # see variable list
dt = 20.0
radius = 400.0 # radii = 300.0 400.0 ...
```

■ SETUP

Output time: 10M or 20M (the latter might already be too large)

Resolution: $(\theta, \phi)_{\text{low}} = (128, 256)$ or $(\theta, \phi)_{\text{high}} = (256, 512)$

Radii: Between 300M and 500M

■ DISK USAGE

- Low resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 30 \times 128 \times 256 \times 8 \approx 7.86\text{MB}(r\Delta t)^{-1}$
- High resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 30 \times 256 \times 512 \times 8 \approx 31.46\text{MB}(r\Delta t)^{-1}$

	10M	20M
• Total (100ms or 20000M):		
	Low resolution	High resolution
	$\approx 15.72 \text{ GBr}^{-1}$	$\approx 62.92 \text{ GBr}^{-1}$
	$\approx 7.86 \text{ GBr}^{-1}$	$\approx 31.46 \text{ GBr}^{-1}$

■ **NOTE** To save disk-usage, this can be paired with the output from **Ejecta** and also **GRMHD butterfly diagrams**.

GRMHD butterfly diagrams

■ VARIABLES

Primitives: `bcc1`, `bcc2`, `bcc3`, `velx`, `vely`, `velz`

Gauge: `z4c_alpha`, `z4c_betax`, `z4c_betay`, `z4c_betaz`

Metric: `adm_gxx`, `adm_gxy`, `adm_gxz`, `adm_gyy`, `adm_gyz`, `adm_gzz`

Total: $N_{\text{vars}} = 16$

■ OUTPUT TYPE

```
<output3>
file_type = sph
variable = ... # see variable list
dt = 20.0
radius = 400.0 # radii = 300.0 400.0 ...
```

■ SETUP

Output time: 10M or 20M (the latter might already be too large)

Resolution: $(\theta, \phi)_{\text{low}} = (128, 256)$ or $(\theta, \phi)_{\text{high}} = (256, 512)$

Radii: Somewhere in the disk (between 50M and 300M)

■ DISK USAGE

- Low resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 16 \times 128 \times 256 \times 8 \approx 4.19\text{MB}(r\Delta t)^{-1}$
- High resolution - $N_{\text{vars}} \times N_{\theta} \times N_{\phi} \times 8 \text{ bytes} = 16 \times 256 \times 512 \times 8 \approx 16.78\text{MB}(r\Delta t)^{-1}$

	10M	20M
• Total (100ms or 20000M):		
	Low resolution	High resolution
	$\approx 8.38 \text{ GBr}^{-1}$	$\approx 33.56 \text{ GBr}^{-1}$
	$\approx 4.19 \text{ GBr}^{-1}$	$\approx 16.78 \text{ GBr}^{-1}$

■ **NOTE** To save disk-usage, this can be paired with the output from **Ejecta** and also **Neutrino luminosity and average energy density**. The computation is performed via $b_{\bar{\phi}} = \frac{xb_y - yb_x}{\sqrt{x^2 + y^2}}$, where b_x and b_y are the magnetic field components along the x- and y-direction in the fluid frame, and $\bar{\phi}$ denotes the average in the azimuthal direction.

Waveforms

Kinetic/magnetic energy evolution

■ **VARIABLES** None.

■ **OUTPUT TYPE**

```
<output4>
file_type   = hst
dt          = 20.0
data_format = %20.15e
```

For this analysis we require the file `<jobname>.mhd.hst`.

■ **SETUP**

Output time: 5M is a good minimum, but can also be increased to 10M or 20M, or lowered to 1M

■ **DISK USAGE**

- Bytes per line - $N_{\text{byte,column}} \times N_{\text{columns}} = \underbrace{22}_{21 \text{ if unsigned}} \times \underbrace{14}_{\text{for 1 passive scalar}} \approx 294 \text{ Bytes}$

- Total (100ms or 20000M):

	1M	5M	10M	20M
Usage	$\approx 5.88 \text{ MB}$	$\approx 1.18 \text{ MB}$	$\approx 0.588 \text{ MB}$	$\approx 0.294 \text{ MB}$

User history output